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Drinking Water: Need for Purification

- **Surface water** from rivers, lakes and reservoirs cannot be directly used for drinking
 - ❑ Unacceptable turbidity & color, suspended solids, high levels of anions and heavy metals, trace organic contaminants
- **Groundwater (GW)** from aquifers may also require treatment to make it fit for consumption
 - ❑ High arsenic levels in GW causes chronic toxicity
 - ❑ High nitrate and fluoride levels in GW causes problems such as blue baby syndrome and dental & skeletal decay, respectively
 - ❑ Leachate generated from dumping grounds and unlined pits can contaminate GW
- **Pathogens in water**
 - ❑ Disease causing microorganisms reside in the intestine of warm-blooded animals
 - ❑ Contamination of water with sewage

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Source of Drinking water

- Characteristics of a suitable drinking water source is defined by the Central Pollution Control Board (CPCB) in India
- **CPCB Classification:**
 - ❑ Classes A, B, C, D, E and others
 - A: Best Quality, Only disinfection required before drinking
 - C: Requires various types of treatment including disinfection
 - ❑ Defined in terms measurable parameters (Class A):
 - Class A: pH range-6.5-8.5; Dissolved oxygen > 6 mg/L; Count of Coliform Organisms (MPN) < 50 per 100 mL;
 - 5-day BOD at 20°C < 2 mg/L

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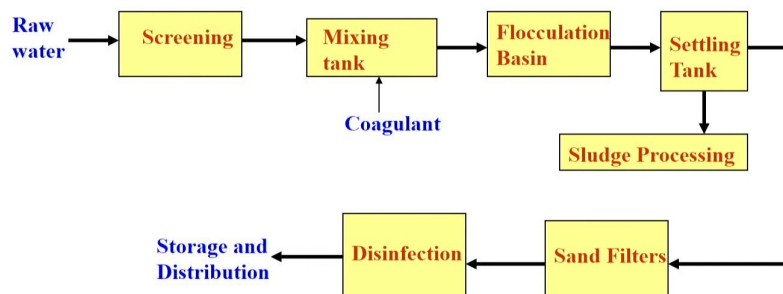
Is Your Tap Water Safe for Drinking ?

- Test if water quality parameters are within acceptable limits specified by authorized agencies
- **Indian Agencies**
 - ICMR: Indian council of medical research
 - BIS: Bureau of Indian standards
 - CPCB: Central & State Pollution Control Boards formulates MINAS (Minimal National standards)
- **International Agencies**
 - WHO: World Health Organisation
 - US- EPA: Environment Protection Agency, USA

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Conventional Water Treatment

Raw water from lake / river is treated to meet the drinking water quality criteria using a series of unit operations



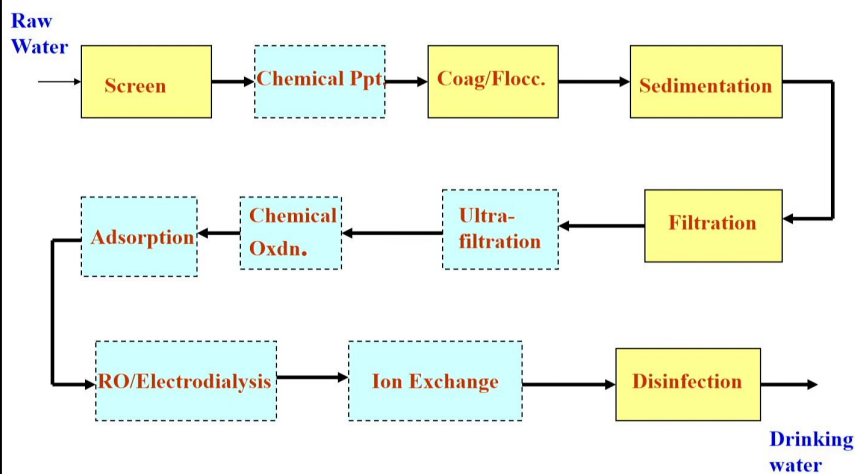
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Conventional Treatment—Unit Operations

- **Screening:** Removes large floating and suspended objects
- **Coagulation:** Small colloidal solids are converted into large suspended particles by adding external agent
- **Flocculation:** Floc formation from destabilized colloidal particles by gentle mixing
- **Settling Tank:** Settling of flocs by gravity
- **Sludge Processing:** Processing of sludge / settled flocs
- **Filtration:** Removes suspended solids
- **Disinfection:** Removal of Pathogens

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Advanced Treatment Process



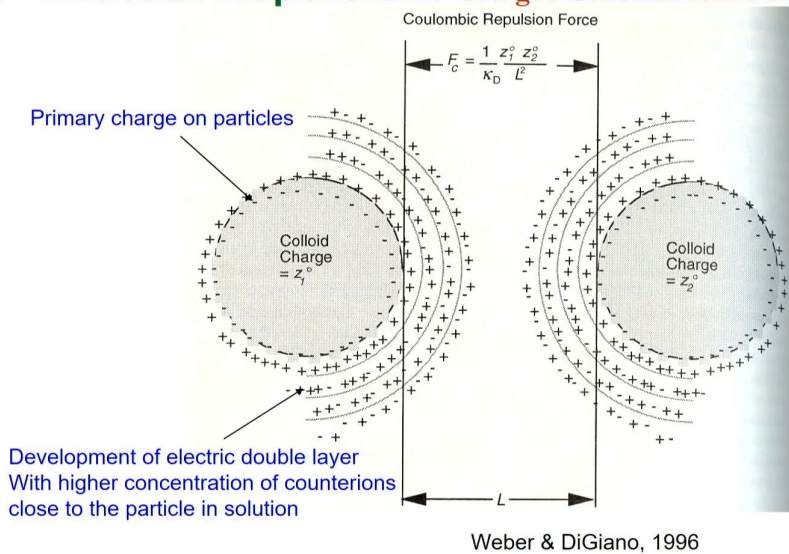
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Coagulation & Flocculation

- Process for removal of colloids and color causing macromolecules
 - **Coagulation:** The chemical process of destabilization
 - Rapid mixing for a short period generates active species that achieves destabilization
 - **Flocculation:** The physical process of transport of particles to form flocs that can settle out, gentle mixing to promote contact
- **Colloids in water**
 - Size range: 0.001-1 μm ; large surface to volume ratio
 - color causing macromolecules, clays, bacteria, viruses, proteins
 - Typically, negatively charged: like charges repel each other, hence colloidal particles resist aggregation

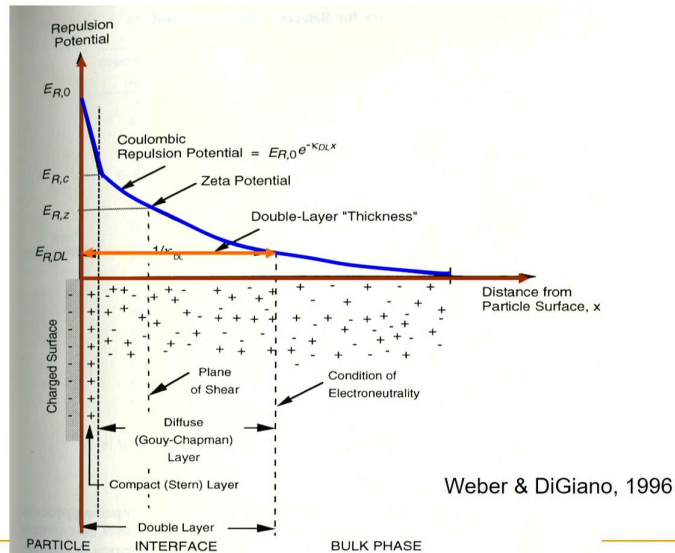
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Mutual Repulsion: Charged Colloidal Particles



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Double Layer Representation



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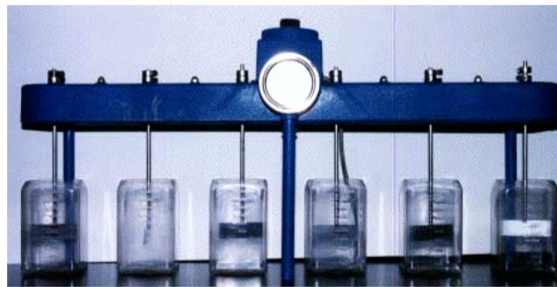
Coagulants & Mechanisms of Coagulation

- **Examples of Coagulants**
 - ❑ Alum ($Al_2(SO_4)_3$), Ferric chloride, Ferrous sulphate, polyelectrolytes
 - ❑ Polyelectrolytes can also be added as coagulant aids
- **Mechanisms of Coagulation**
 - ❑ Double Layer Compression by indifferent electrolyte
 - ❑ Adsorption and Charge Neutralization by potential determining electrolyte
 - ❑ Enmeshment in a precipitate
 - ❑ Adsorption and interparticle bridging by polymers-nonionic / anionic / cationic
- **A coagulant such as alum can act by more than one mechanism**
 - ❑ The dominant mechanism is affected by pH, alkalinity and colloid concentration

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Jar Test Apparatus

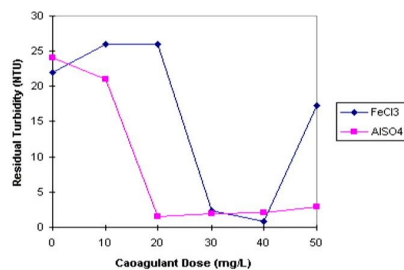
Coagulation-Flocculation and Settling



- ✓ Six Containers → For testing various coagulant dose + 1 Control
- ✓ Variable mixing speed, same for all containers
- ✓ 100 rpm 1 min; 25-35 rpm 20-30 min; no mixing 1 h

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Jar Test: Turbidity Vs Coagulant Dose



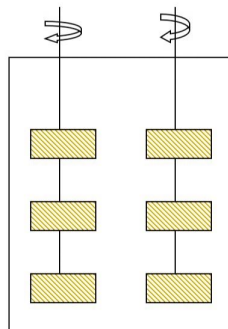
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Flocculation

- The rate of aggregation of destabilized colloidal particles; depends on:
 - Rate at which collision occurs between particles
 - Effectiveness of these collisions in causing attachment
- Mechanisms of interparticle contact
 - Contacts by thermal motion (Brownian motion), bulk fluid motion (Induced by stirring), & particle settling
- Two types of flocculation processes
 - **Perikinetic flocculation** → Intrinsic property controlled by Brownian motion
 - **Orthokinetic flocculation** → Affected by mean velocity gradients ($\bar{G}$) characterizing spatial changes in velocity; relates to power input for mixing
 - Typical $\bar{G}$ value for flocculation: $10\text{-}100\text{ s}^{-1}$ and retention time in flocculation basin 20-60 min; Gt value determines the mass of flocs formed

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Flocculation

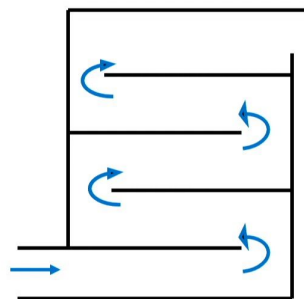


Paddle Flocculator

- ✓ Gentle agitation promotes contact between destabilized particles

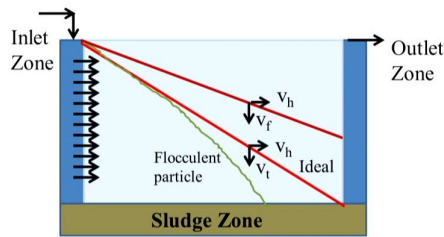
Baffled Flocculator

- ✓ Long flow path promotes contact between destabilized particles



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Discrete Particle Settling



The treated water is skimmed from the surface; the sludge containing the settled solids is collected at the bottom in sludge hoppers

$$t_D = \frac{V}{Q} = \frac{L.W.D}{Q} = \frac{D}{v_t}$$

$$v_t = \frac{Q}{L.W} = \frac{Q}{A_s} = \text{Overflow rate}$$

✓ Particles that completely settle are determined not by depth (D) but by the tank surface area (A_s)

✓ For discrete particle $v_t = f(d_p)$ is the terminal settling velocity given by Stokes law

t_D is the detention time; v_t is the terminal settling velocity

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Class 2 Settling: Settling of Flocculant particles

- ✓ Settling of discrete particles occurs as per class 1 settling behavior; Settling of flocculant particles is class 2 settling
- ✓ Particles coalesce / flocculate during sedimentation, hence the mass increases and the particles settle faster
- ✓ Settling efficiency is determined through sedimentation tests in settling column
- ✓ Extent of flocculation is a function of contact opportunities, overflow rate, depth, velocity gradients, particle conc. and particle size range

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Rapid Sand Filtration

- Removes the flocs that resist settling; they can hinder disinfection
- Consists of a bed of sand packed in a deep bed above a layer of gravel
- The removal of particulate materials by accumulation throughout the depth of a filter medium as water passes through it. NTU at effluent ~0.5 NTU

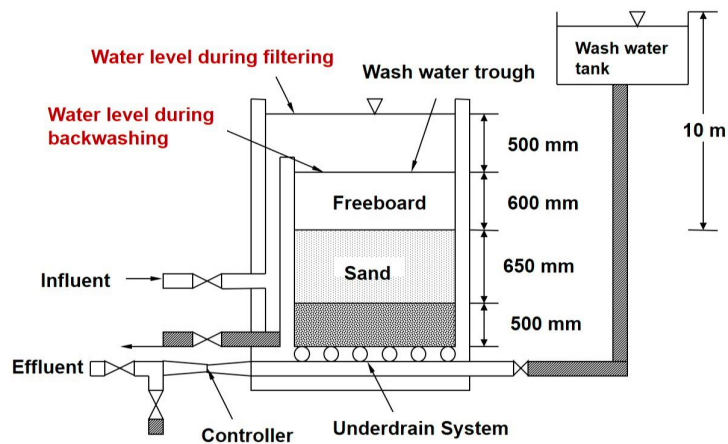
- Primary Mechanism: Physicochemical filtration**

particle removal by physical and chemical forces between the particles (p) and the media/collector (c) – transport and attachment of particulates ($d_c/d_p \approx 1000$)

Little decrease in permeability compared to cake filtration/straining filtration

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Rapid Sand Filter



Typical Effective size of sand > 0.45 mm; Uniformity Coefficient < 1.5

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Rapid Sand Filtration

- Filter media: bed of sand grains called collectors of a suitable size range
 - Effective size d_{10} :size of grains in mm such that 10% of particles by weight are smaller
 - Uniformity coefficient $C_u = d_{60}/d_{10}$
- Consists of 0.45-0.76 m of sand supported by 0.45 m of gravel. d_{10} range 0.45-0.7 mm. C_u range 1.2-1.75
- Flow rate range 1- 4 gpm/sq ft (2.4-34 m/h)
- Can be operated as gravity filters or pressure filters
- Cleaned by backwashing

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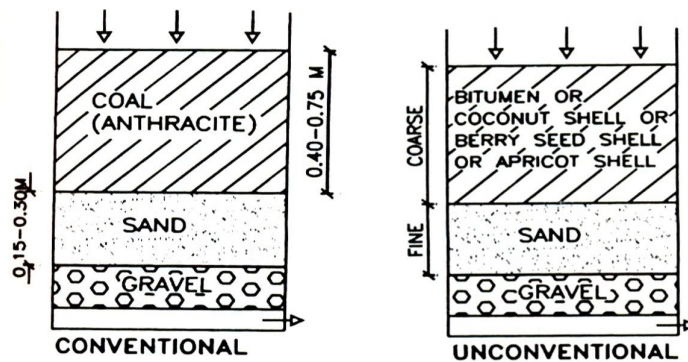
Backwashing

- With time, deposits build up, decreasing porosity and permeability of filter media
- Head loss increases, flow rate decreases at a constant hydraulic head
- Shear stress on deposited particles increases and water quality decreases
- Backwashing involves flow reversal through the bed. Hydraulic shear dislodges particles. Also includes surface scour
- Typical flow rates 36-48 m/h. Time of backwashing 3-15 min after a 12 hr run. Volume of backwash water 2-5% of filtered water. Accompanied by bed expansion of 20-50%
- For media particles with similar specific gravity, backwashing causes fine to coarse grading of the filter media with increasing depth

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Dual Media/ Multi-Media Filters

- A special type of rapid sand filter, i.e., collectors of different material and specific gravity
- Coarse grains of a lighter material can lie above finer grains of a heavier material.



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Slow Sand Filters

- Finer sand grains (0.2 mm) of greater (0.9-1.5m) depth; lower filtration rate 0.1-0.3 m/h
- Produces good water quality but requires large land area
- Solids removal in upper zone "Schmutzdecke" accompanied by some biological activity. Cleaning by scrapping the top 1" at 1-6 months interval.

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Disinfection Using Chlorine

- Scope: To destroy the pathogens
 - Common Disinfectants:
 - Cl_2 gas (most widely used)
 - Calcium hypochlorite $\text{Ca}(\text{OCl})_2$
 - Sodium hypochlorite $\text{Na}(\text{OCl})_2$
 - Ozone
 - UV Radiation
- $$\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HOCl} + \text{H}^+ + \text{Cl}^-$$
- $$\text{HOCl} \leftrightarrow \text{H}^+ + \text{OCl}^-$$
- Residual Chlorine $\rightarrow$ Chlorine available for disinfection
 - Free available residual chlorine $\rightarrow [\text{HOCl}] + [\text{OCl}^-]$

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Disinfection Using Chlorine

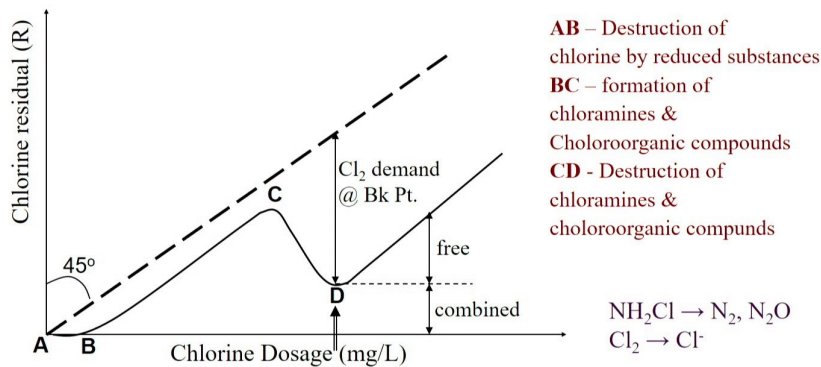
- **Chlorine demand**
 - Exerted by organic and inorganic substances in reduced form that are oxidized by chlorine
 - eg. Inorganic compounds : Fe^{+2} , Mn^{+2} , NO_2^- , H_2S
 - eg. Organic compounds: Ammonia, Organic matter.
 - **Reactions of NH_3 with Chlorine**
 - NH_3 often present in wastewater
 - Mono, di and tri chloramines are formed
 - Chloramines have germicidal action but they are slow acting
- $$\text{NH}_3 + \text{HOCl} \rightarrow \text{NH}_2\text{Cl} + \text{H}_2\text{O}$$
- $$\text{NH}_2\text{Cl} + \text{HOCl} \rightarrow \text{NHCl}_2 + \text{H}_2\text{O}$$
- $$\text{NHCl}_2 + \text{HOCl} \rightarrow \text{NCl}_3 + \text{H}_2\text{O}$$
- $[\text{NH}_2\text{Cl}] \text{ \& } [\text{NHCl}_2] = f(\text{pH, temperature, Contact time, HOCl})$

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Disinfection Using Chlorine

■ Combined Residual Chlorine

- Chlorine residual in combination with NH_3 and organic nitrogen compounds



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Disinfection Using Chlorine

■ Breakpoint Chlorination:

- Variation of residual chlorine in water with increasing Cl_2 dose for water and wastewater containing ammonia and other organic nitrogen compounds
- Cl_2 dosage beyond breakpoint will cause a proportional increase in free available chlorine
- Depending on water quality, chlorine dose up to 10 mg/L may be needed to obtain chlorine residual of 0.5 mg/L

- For good chlorination use Cl_2 dose > Cl_2 at breakpoint

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Bacterial/Germicidal action of Cl_2

- Oxidize the bacterial cell's chemical structure thereby destroying enzymatic process
- Protozoan cysts & enterovirus are more resistant to Cl_2
- Rate of disinfection = f (conc. & form of available residual chlorine, pH, Temperature, contact time)
 - $\text{HOCl} \rightarrow$ Hypochlorous acid is 40-80% most effective – can kill virus
 - $\text{OCl}^- \rightarrow$ Hypochlorite ions are less effective
- Chloramines $\rightarrow$ disinfecting power lower than OCl^- , more persistent
- Some combined residual is desirable (although less effective)
 - Protect DW recontamination in distribution system
 - Controls algal & bacterial after-growth in treated waters
- Problem of Cl_2 : Conversion of organic matter to carcinogenic Trihalomethanes (THMs)

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Disinfection Kinetics

Chick's Law N is the number concentration of microorganisms
 $N_0 = N$ at $t=0$; k is a constant

$$\frac{dN}{dt} = -kN$$

$$\ln\left(\frac{N}{N_0}\right) = -kt$$

Modified Chick's Law

$$\ln\left(\frac{N}{N_0}\right) = -kt^m$$

$$\ln\left(-\ln\left(\frac{N}{N_0}\right)\right) = \ln k + m \ln t$$

m & k are functions of type & concentration (C) of chemical agent, temperature, type of organism, composition of the medium

$m = 1$ Chick's Law

$m < 1$ Rate of kill decrease with time

$m > 1$ Rate of kill increase with time

Dependence on Residual chlorine concentration

$$N/N_0 = (1 + 0.23Ct)^{-3}$$

Valid in a Batch Reactor

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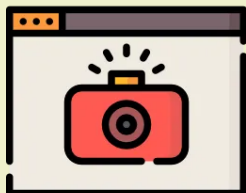
Disinfection

- Disinfection of water containing natural organic matter (NOM) with chlorine may cause generation of disinfection by-products (DBPs) such as trihalomethanes (THMs)
- Apart from the disinfection process other processes also causes removal of pathogens, such as protozoan cysts
 - Coagulation-Flocculation & Settling
 - Filtration
 - Softening with Lime Soda process (at high pH)
- Removal of turbidity and NOM is essential for achieving good disinfection

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